

Notation. Let $\alpha: I \rightarrow \mathbb{R}^3; t \mapsto (\alpha_1(t), \alpha_2(t), \alpha_3(t))$ be a regular Curve.

$$\frac{d\alpha}{dt} = \frac{d}{dt} \alpha(t) = \left(\frac{d}{dt} \alpha_1(t), \frac{d}{dt} \alpha_2(t), \frac{d}{dt} \alpha_3(t) \right).$$

Re Parametrization

Given a regular Curve $\alpha: I \rightarrow \mathbb{R}^3$, generally, $|\frac{d\alpha}{dt}| \neq 1$.

But, Consider

$$S = S(t) = \int_{t_0}^t \left| \frac{d\alpha}{du} \right| du. \quad \text{Then, } \frac{dS}{dt} = \left| \frac{d\alpha}{dt} \right|$$

$$\text{Now, } \frac{d\alpha}{dS} = \frac{d\alpha}{dt} \cdot \frac{dt}{dS} = \frac{d\alpha}{dt} \cdot \left| \frac{d\alpha}{dt} \right|^{-1}.$$

